Module 3 — Quiz A (Def & Notation)

Question bank · After Read 1 · open notes · unlimited attempts

Module 3, Quiz A: Definitions & notation

Reading: Module 3 Read 1
Project: Project 3 · **Canvas:** Def & Notation quiz (Module 3)

Work these **practice** items after Read 1. Sections cover: tidyverse pipeline verbs, variable type classification, and spread/IQR vocabulary from Read 1.

Graph application and two-way table reading → Quiz B after Read 2.

Section A — Tidyverse pipeline verbs (Read 1 · pipeline)

A1. Pipe operator

What is the main purpose of `%>%` (pipe) in the tidyverse?

A2. filter()

Which tidyverse function **keeps rows** that satisfy a logical condition (e.g. `weight > 50`)?

A3. select()

Which function **chooses columns** by name?

A4. mutate()

Which function **adds a new column** while keeping the original number of rows?

A5. count()

Which function counts **how many rows** fall in each level of a categorical variable?

A6. group_by() + summarize()

What is the role of `group_by(species)` before `summarize(mean_mass = mean(body_mass_g))`?

A7. summarize()

After `group_by(island)`, you call `summarize(n = n(), avg = mean(bill_length_mm))`. What do you get?

A8. glimpse()

Name two things `glimpse(df)` shows.

Answer key — Section A — Tidyverse pipeline verbs (Read 1 · pipeline)

1. The pipe passes the result of the left side as the **first argument** to the function on the right — it chains steps so you can read the pipeline top to bottom.
2. `filter()` — it returns only rows where the condition is `TRUE`.
3. `select()` — e.g. `select(species, body_mass_g)` returns a data frame with only those columns.
4. `mutate()` — e.g. `mutate(mass_kg = body_mass_g / 1000)` creates a new column.
5. `count(variable)` — returns a data frame with one row per level and a column `n` with the count.
6. `group_by()` splits the data by `species` so `summarize()` returns one summary row **per species** instead of one row for the whole dataset.
7. One row per island with two columns: `n` (count of penguins on that island) and `avg` (mean bill length for that island).
8. Variable **names**, **types** (e.g. `<dbl>`, `<chr>`), total rows and columns, and a short preview of values.

Section B — Variable types and one-variable summaries

B1. Numerical vs categorical

Classify each as **numerical** or **categorical** (and **binary** if categorical):

a) Penguin flipper length (mm)

b) Island: Biscoe / Dream / Torgersen

c) Sex: male/female

d) Clutch completion: yes/no

e) Bill depth (mm)

B2. Best summary for one numerical variable

For **one** numerical variable, name one graph and two numeric summaries.

B3. Best summary for one categorical variable

For **one** categorical variable, name one graph and one numeric summary.

B4. When to prefer median over mean

Why might you report the **median** instead of the **mean** for a numerical variable?

Answer key — Section B — Variable types and one-variable summaries

1. a) Numerical. b) Categorical, not binary. c) Categorical, binary. d) Categorical, binary. e) Numerical.
2. Graph: **histogram** (or dot plot/boxplot). Numeric: **mean + SD** (symmetric) or **median + IQR** (skewed/outliers).
3. Graph: **bar chart**. Numeric: **counts** or **proportions** per category.
4. The median is **resistant to outliers** — extreme values pull the mean but not the median, so median better represents a 'typical' value when data are skewed or have outliers.

Section C — Spread: range, IQR, and standard deviation

C1. Range

Data: $4, 6, 8, 10, 10, 12, 14, 16$. Compute the **range**.

C2. Quartiles and IQR

Dataset N: $7, 8, 9, 10, 10, 11, 12, 13$. Find Q_1, Q_3, and **IQR**.

C3. Five-number summary

List the five numbers in the **five-number summary** in order.

C4. What a boxplot shows

Describe how a **boxplot** represents the five-number summary.

C5. SD interpretation

In words, what does the **sample standard deviation** s measure?

C6. IQR vs SD

When would you prefer **IQR** over **SD** to describe spread? Why?

Answer key — Section C — Spread: range, IQR, and standard deviation

1. Range $= \max - \min = 16 - 4 = 12$.
2. Lower half: $7, 8, 9, 10 \rightarrow Q_1 = (8+9)/2 = 8.5$. Upper half: $10, 11, 12, 13 \rightarrow Q_3 = (11+12)/2 = 11.5$. IQR $= 3$.
3. Minimum, Q_1 (first quartile), Median (Q_2), Q_3 (third quartile), Maximum.
4. The **box** spans Q_1 to Q_3; the **line inside** is the median; **whiskers** extend to the min and max (or fences if outliers are shown as dots).
5. s measures the typical distance of individual observations from their mean — a larger s means values are more spread out from $\bar{x}$.
6. When the data are **skewed** or have **outliers** — IQR is based only on the middle 50% and is resistant to extremes, while SD uses all values and is sensitive to outliers.
